

Project Proposal: 3D Reconstruction from Images

1. Introduction

1.1 Background

3D reconstruction from images is fundamental in many applications, such as cultural heritage digitalization, industrial reverse engineering, and AR/VR. However, the current 3D reconstruction still has many limitations. The professional 3D reconstruction often requires expensive equipment and time. While in most similar student projects, the images always has no data for the camera poses, which means the algorithm needs to predict the camera poses. Meanwhile, although the NeRF has been used in this field, most of the usage convert a target immediately to an another 2D image (often real-time and from a new angle), don't improve the quality of the reconstruction (3D model).

To address these gaps, developing a low-cost, automated image capture system and integrating it with mature 3D reconstruction tools (e.g., COLMAP) and advanced neural radiance field techniques (e.g., Instant-NGP) for high-precision 3D reconstruction is of significant practical value.

1.2 Project Goals

This project aims to achieve the following objectives:

- Construct an automated image capture hardware system that uses programmable stepper motors to control camera movement (vertical and horizontal directions) and turntable rotation. This system will enable multi-view, uniform, and high-quality image capture of target objects while accurately recording camera pose information.
- Leverage the COLMAP tool to perform 3D reconstruction (from multi-view images to sparse point clouds, dense point clouds, and triangular meshes) using pre-recorded camera pose information (obtained from the hardware system or parsed from image filenames). Additionally, compare the results of two reconstruction approaches—direct dense reconstruction with known camera parameters vs. dense reconstruction based on sparse point clouds—to verify the improvement in reconstruction accuracy brought by using known poses.
- Our advanced goal: If conditions permit, explore the integration of Instant-NGP (a lightweight NeRF implementation) to optimize the details of 3D models generated by COLMAP. Compare the quality of models produced by different techniques and attempt to build an efficient 3D reconstruction workflow: automated capture + COLMAP + Instant-NGP.

2. Related Work (Overview of Core Algorithms)

Below is an introduction to the core algorithm workflows of the key technologies involved in this project:

2.1 COLMAP

COLMAP achieves reconstruction from images to 3D models through two steps: "Structure from Motion (SfM)" and "Multi-View Stereo (MVS)". Its simplified core workflow is as follows:

2.1.1 Structure from Motion (SfM) Algorithm

- Purpose: Estimate camera poses (position + angle) and sparse 3D points of the scene from unordered images
- Core Steps:
 1. Extract feature points from images and match corresponding points between different images;
 2. Filter valid information from the matched points to obtain the relative pose (rotation + translation) of two images;
 3. Calculate the 3D coordinates of matched points using "triangulation" based on the poses, generating an initial sparse point cloud;
 4. Gradually add new images, use the existing point cloud to estimate the pose of new images and expand the point cloud, and perform regular optimization to reduce errors.

2.1.2 Multi-View Stereo (MVS) Algorithm

- Purpose: Complement the sparse point cloud into a dense 3D point cloud, then generate a mesh model
- Core Steps:
 1. Combine adjacent view images to calculate the pixel depth value of each image, generating a single-view depth map;
 2. Convert all depth maps to the same coordinate system, fuse and synchronize them while eliminating conflicting data, obtaining a dense point cloud;
 3. Convert the dense point cloud into a triangular mesh, map textures from the original images, and get a textured 3D model.

2.2 NeRF

NeRF uses a neural network to "implicitly encode the scene" and generates 3D structures through "volume rendering". Its core workflow is as follows:

1. Radiance Field Modeling:

- Represent the scene as a "radiance field" (color + density of each point in space);
- Use an 8-layer fully connected neural network (MLP) to learn the radiance field: input "spatial coordinates + viewing direction", output "color + density of the point";
- Use "positional encoding" to map coordinates/viewing directions to a high-dimensional space, helping the network learn details.

2. Image Generation via Volume Rendering:

- Simulate the process of camera rays passing through the scene: for each ray, sample multiple points within the scene (within the effective range from near to far);
- Calculate the "contribution value" of each sampled point: density determines whether the point is a solid (density > 0 means a solid), color determines the point's appearance, and at the same time, combine "transmittance" (the probability that the ray is not absorbed from the starting point to this point, related to the cumulative density of all points along the path) to comprehensively obtain the point's contribution to the ray's color;
- Accumulate the contribution values of all sampled points to get the final color of the ray; repeatedly calculate all rays to generate an image of the corresponding view, while completing the 3D reconstruction of the scene.

2.3 Instant-NGP

Instant-NGP is an optimized version of NeRF, with the core goal of "improving efficiency". Its key algorithm improvements are as follows:

1. Multi-Resolution Hash Encoding:

- Replace NeRF's high-dimensional positional encoding: divide the scene into 3D grids of multiple resolutions (from low to high resolution), with each grid corresponding to a short feature vector;
- Use a hash table to store these feature vectors (greatly reducing storage capacity); for any point in space, fuse its features in grids of various resolutions through interpolation, and use this as the network input, significantly reducing the amount of computation.

2. Efficiency Optimization:

- Supports direct initialization using camera poses output by COLMAP, no manual annotation required;
- Uses a "spatial partitioning tree" to quickly locate solid regions in the scene, reducing sampling of invalid regions (such as air) and further lowering computational costs;
- Based on GPU parallel computing, it achieves "minute-level training" (only 5-10 minutes for small scenes) and "real-time rendering" (frame rate up to 30fps or higher).

3. Our Approach

3.1 Hardware Framework: Automated Capture System

In the previous lab, we already know that we can use python to capture images automatically. To capture images from various angles, we have two ways:

1. Use a camera which can move up and down, and a rotary table, to help us capture images from various angles at a particular height.
2. Use a camera which can move up and down and left and right, this may help us capture images in a more natural light field.

The programmable stepper motor is required for the motion structure. Additionally, an LED ring fill light may be used to enhance the light and avoid the influence of shadows.

3.2 Software Framework: COLMAP

Step 1: Image Capture

Capture images from various angles using the automated capture system, and record the camera poses.

Step 2: Image Processing

Process the captured images and generate the parameter files `cameras.txt` and `images.txt`. What's more, we will use COLMAP's built-in camera calibration to remove lens distortion (via a chessboard pattern). Normalize lighting with CLAHE, and then estimate camera poses and generate a sparse point cloud.

Step 3: 3D Reconstruction

On the one hand, import two parameter files into COLMAP, perform dense reconstruction directly, and obtain the dense point cloud and triangular mesh output (OUTPUT_1).

On the other hand, use the sparse point cloud to get the dense point cloud and triangular mesh output (OUTPUT_2).

Compare the quality of the outputs from the two methods (OUTPUT_1 and OUTPUT_2).

3.3 Advanced Approach: Instant-NGP

Instant-NGP is a lightweight NeRF implementation that can be used for 3D reconstruction. If conditions permit, we will try to run Instant-NGP locally to get the detailed 3D reconstruction

results(OUTPUT_3) and compare it with the outputs from COLMAP(OUTPUT_1 and OUTPUT_2). And combine the Instant-NGP and COLMAP to get a matured 3D reconstruction work flow.

4. Milestones(4 Weeks)

Week	Task
5	Setup the automated capture system, setup the software framework, and start the project.
6	Tune the automated capture system, and write the image processing code, test the COLMAP.
7	Continue to adjust the COLMAP, and start to study NeRF and the Instant-NGP.
8	Combine the COLMAP and Instant-NGP if possible, and write the document.

5. References

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